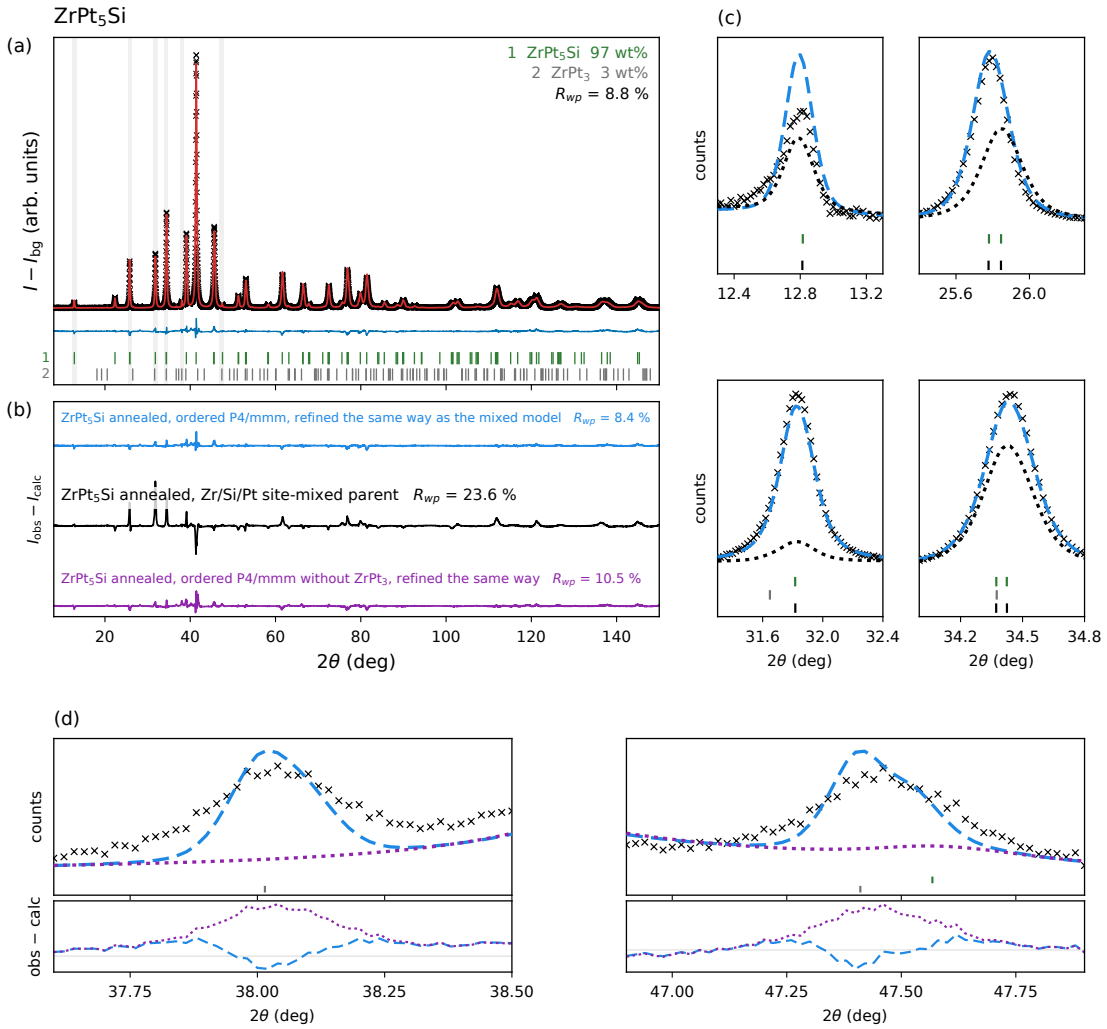
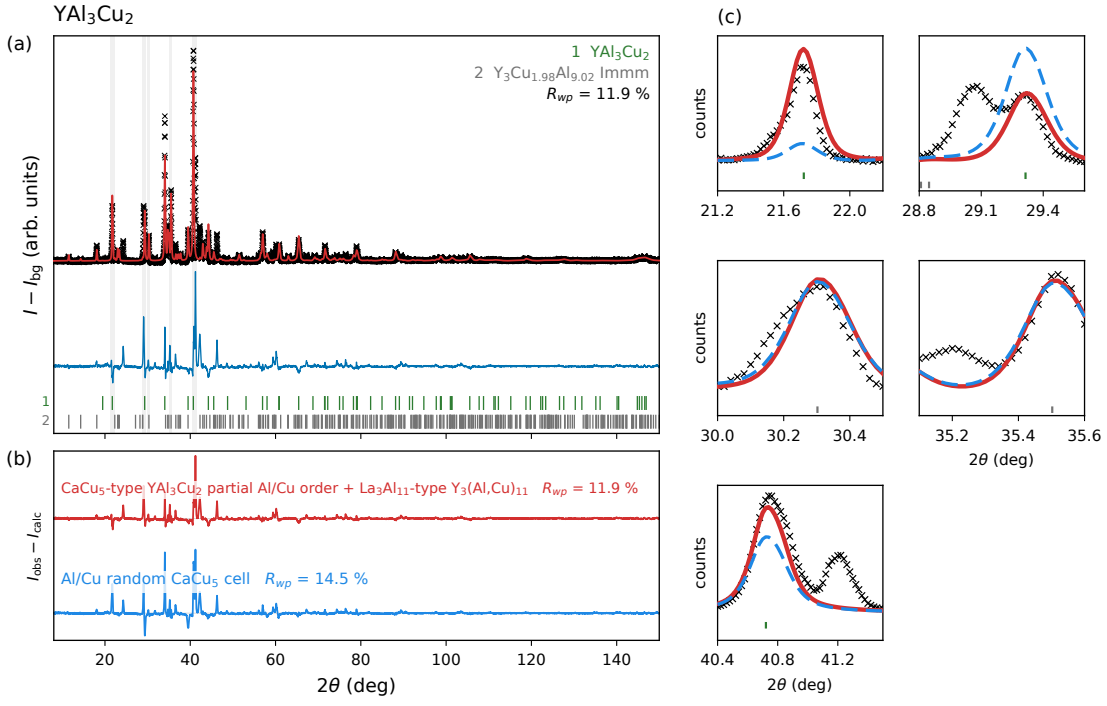
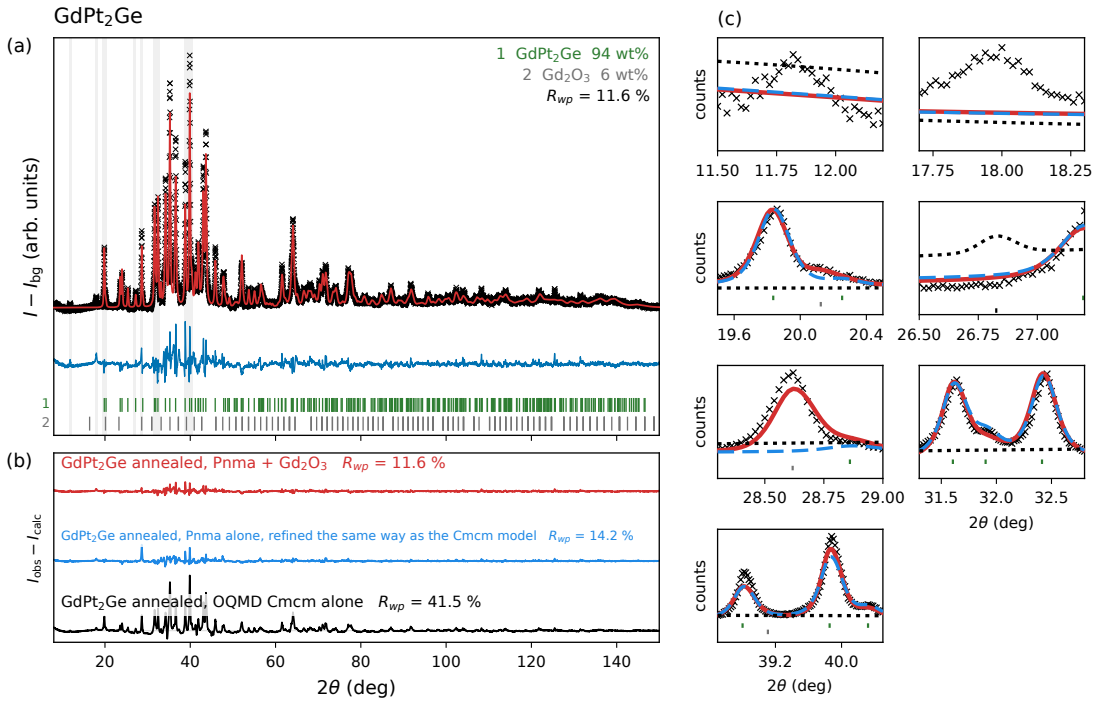
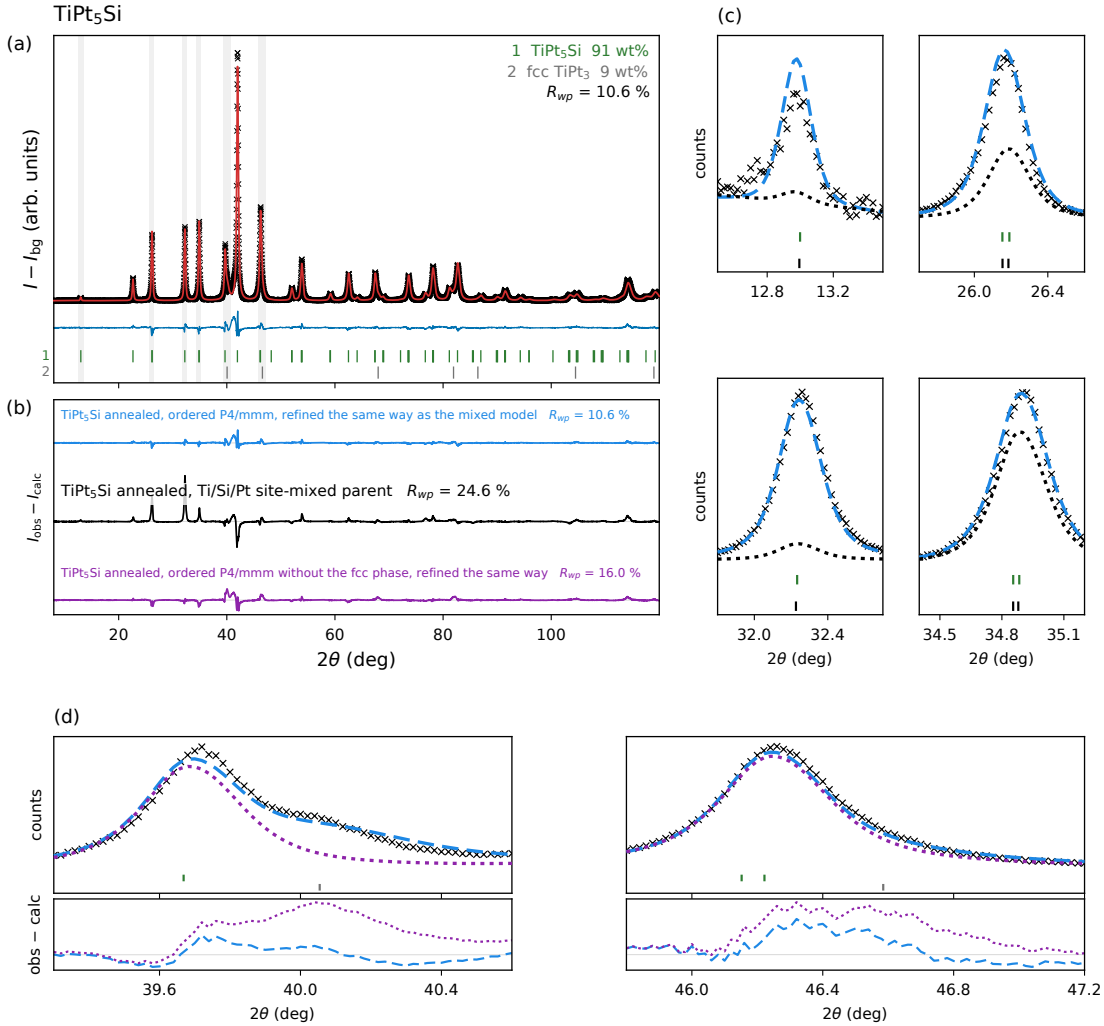
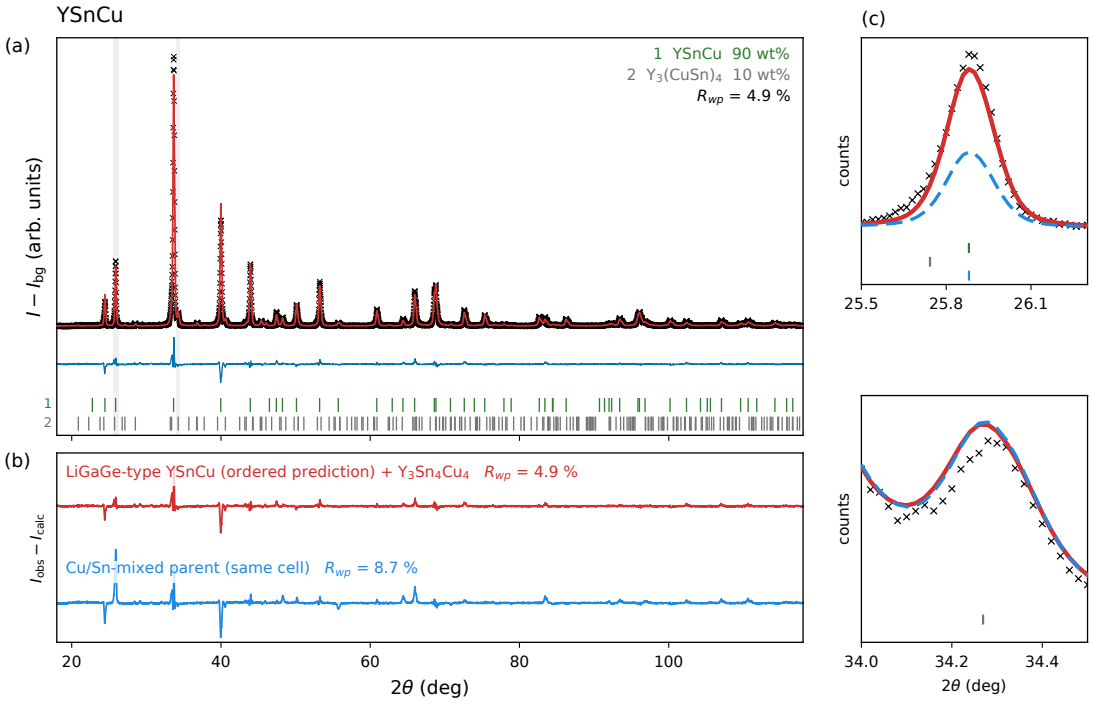
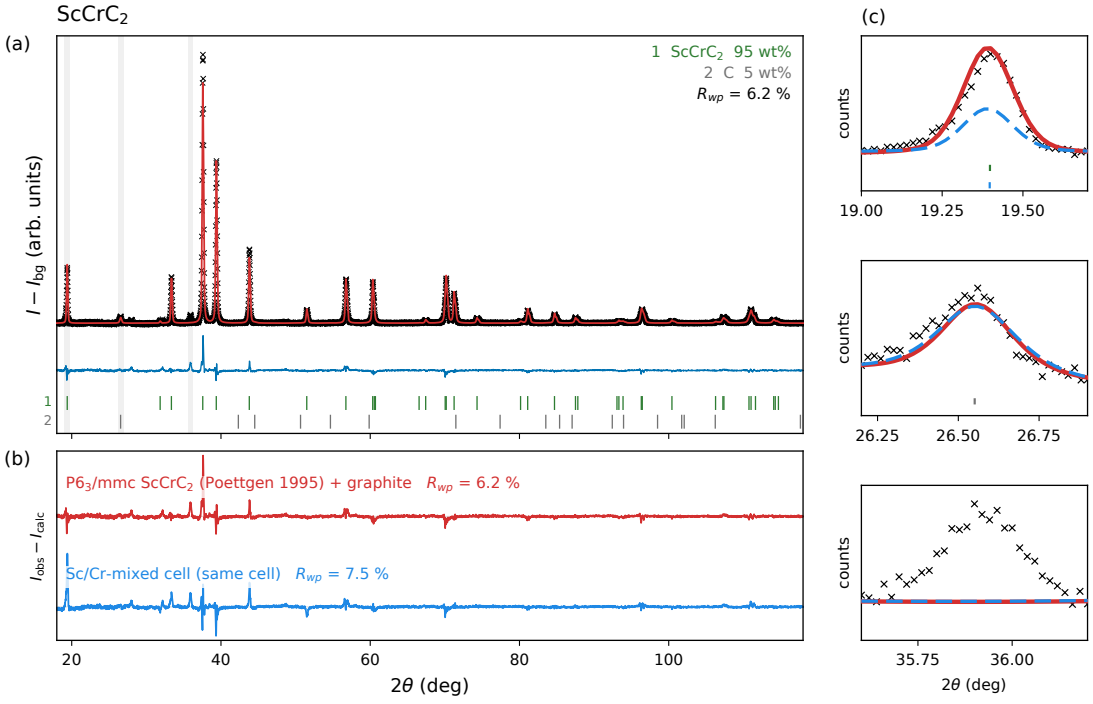

ZrPt5Si
(a)
1 ZrPt5Si 97 wt%
2 ZrPt3 3 wt%
Rwp = 8.8 %
I − Ibg (arb. units)
(b)
ZrPt5Si annealed, ordered P4/mmm, refined the same way as the mixed model Rwp = 8.4 %
ZrPt5Si annealed, Zr/Si/Pt site-mixed parent Rwp = 23.6 %
ZrPt5Si annealed, ordered P4/mmm without ZrPt3, refined the same way Rwp = 10.5 %
Iobs − Icalc
2θ (deg)
(c)
counts
(d)
obs − calc

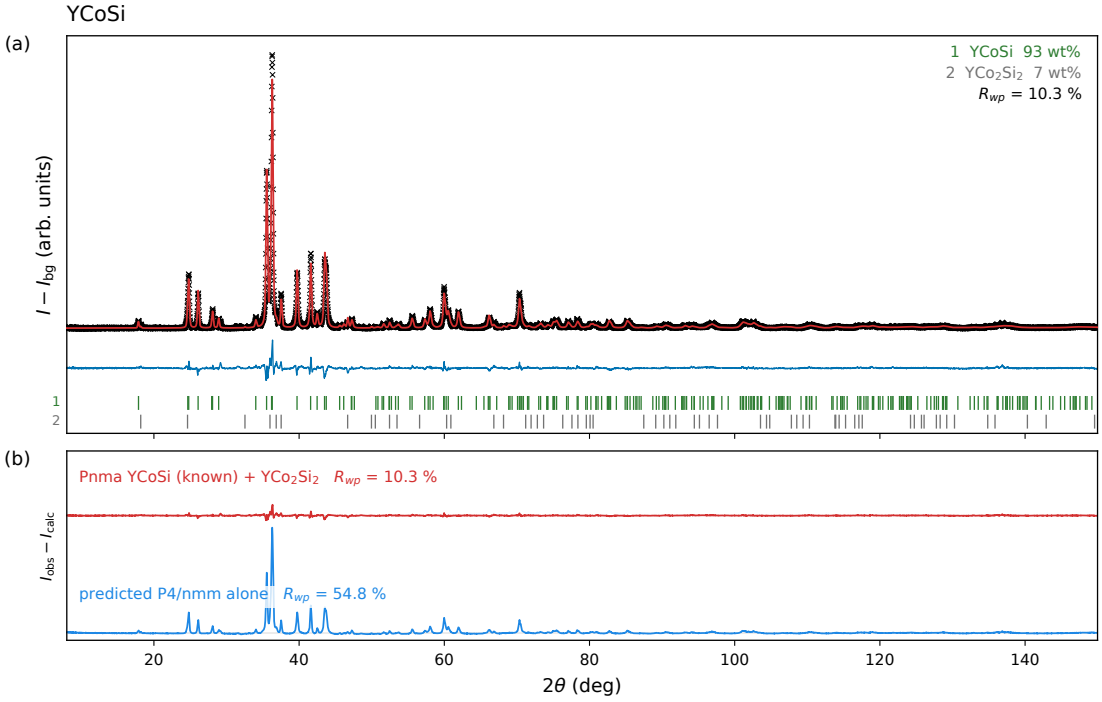

YAl3Cu2
(a)
1 YAl3Cu2
2 Y3Cu1.98Al9.02 Immm
Rwp = 11.9 %
I − Ibg (arb. units)
(b)
CaCu5-type YAl3Cu2 partial Al/Cu order + La3Al11-type Y3(Al,Cu)11 Rwp = 11.9 %
Al/Cu random CaCu5 cell Rwp = 14.5 %
Iobs − Icalc
2θ (deg)
(c)
counts

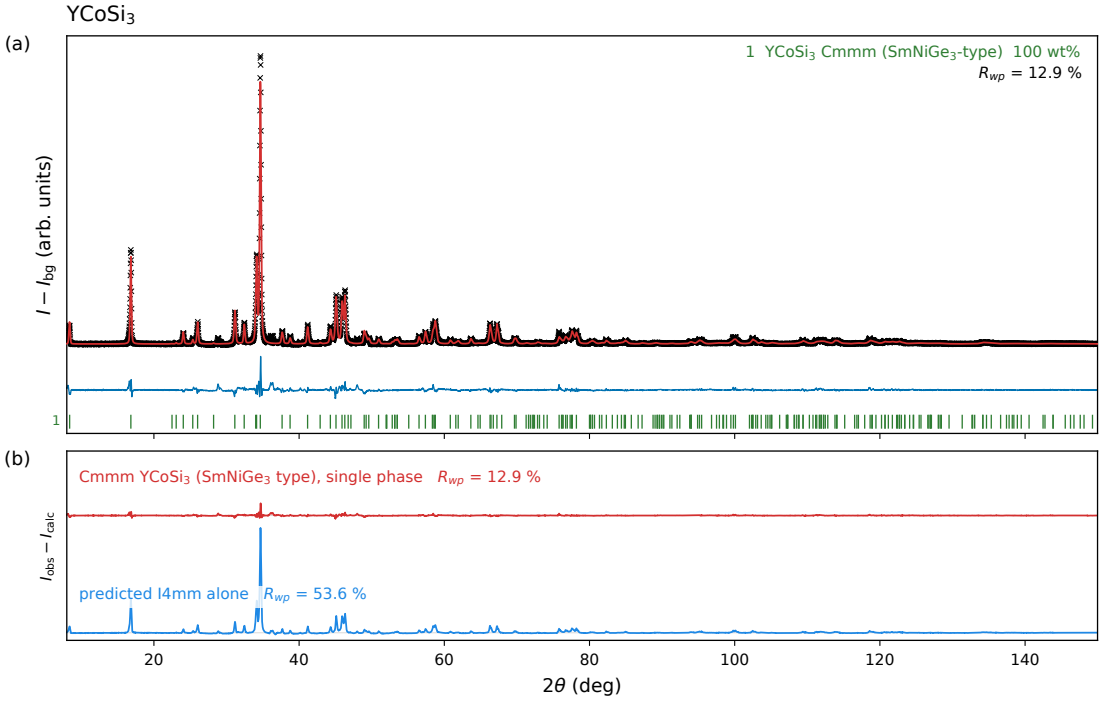

GdPt2Ge
(a)
1 GdPt2Ge 94 wt%
2 Gd2O3 6 wt%
Rwp = 11.6 %
I − Ibg (arb. units)
(b)
GdPt2Ge annealed, Pnma + Gd2O3 Rwp = 11.6 %
GdPt2Ge annealed, Pnma alone, refined the same way as the Cmcm model Rwp = 14.2 %
GdPt2Ge annealed, OQMD Cmcm alone Rwp = 41.5 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)

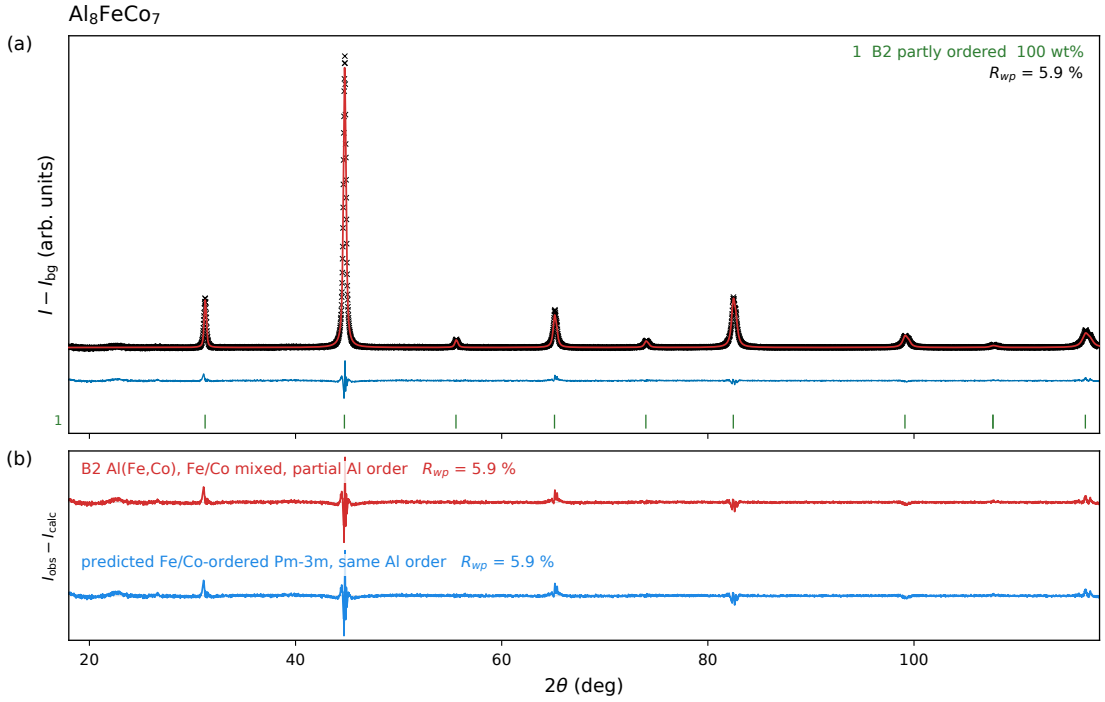

TiPt5Si
(a)
1 TiPt5Si 91 wt%
2 fcc TiPt3 9 wt%
Rwp = 10.6 %
I − Ibg (arb. units)
(b)
TiPt5Si annealed, ordered P4/mmm, refined the same way as the mixed model Rwp = 10.6 %
TiPt5Si annealed, Ti/Si/Pt site-mixed parent Rwp = 24.6 %
TiPt5Si annealed, ordered P4/mmm without the fcc phase, refined the same way Rwp = 16.0 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)
(d)
counts
obs − calc
2θ (deg)

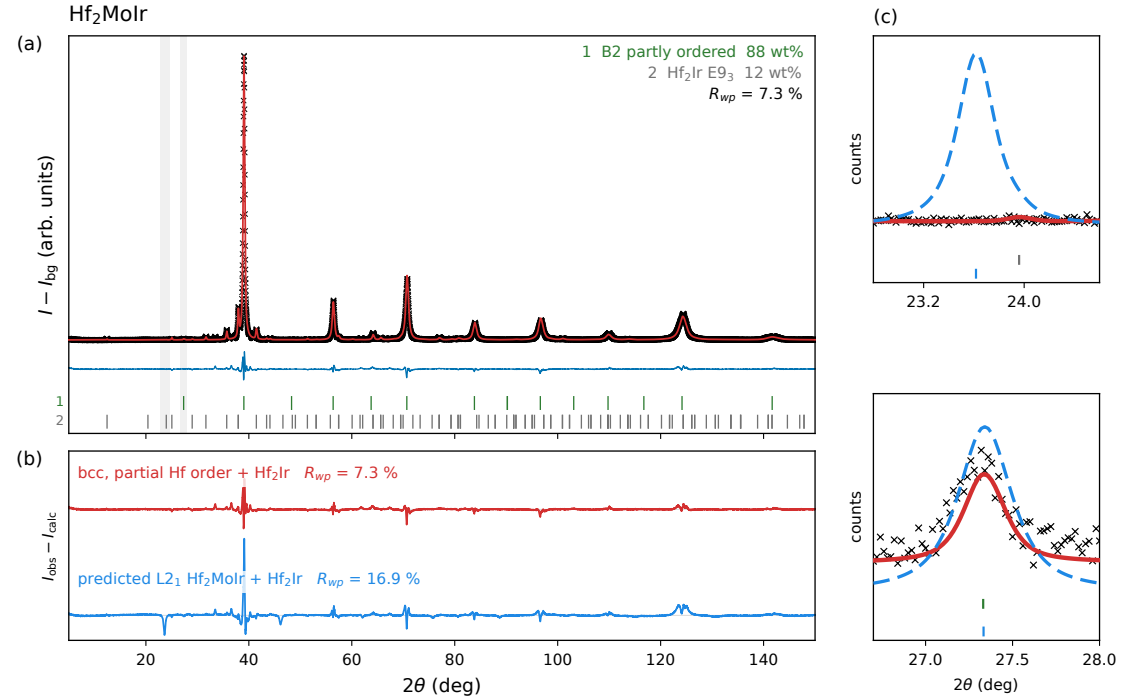

YSnCu
(a)
1 YSnCu 90 wt%
2 Y3(CuSn)4 10 wt%
Rwp = 4.9 %
I − Ibg (arb. units)
1
2
(b)
LiGaGe-type YSnCu (ordered prediction) + Y3Sn4Cu4 Rwp = 4.9 %
Cu/Sn-mixed parent (same cell) Rwp = 8.7 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
25.5
25.8
26.1
34.0
34.2
34.4
2θ (deg)

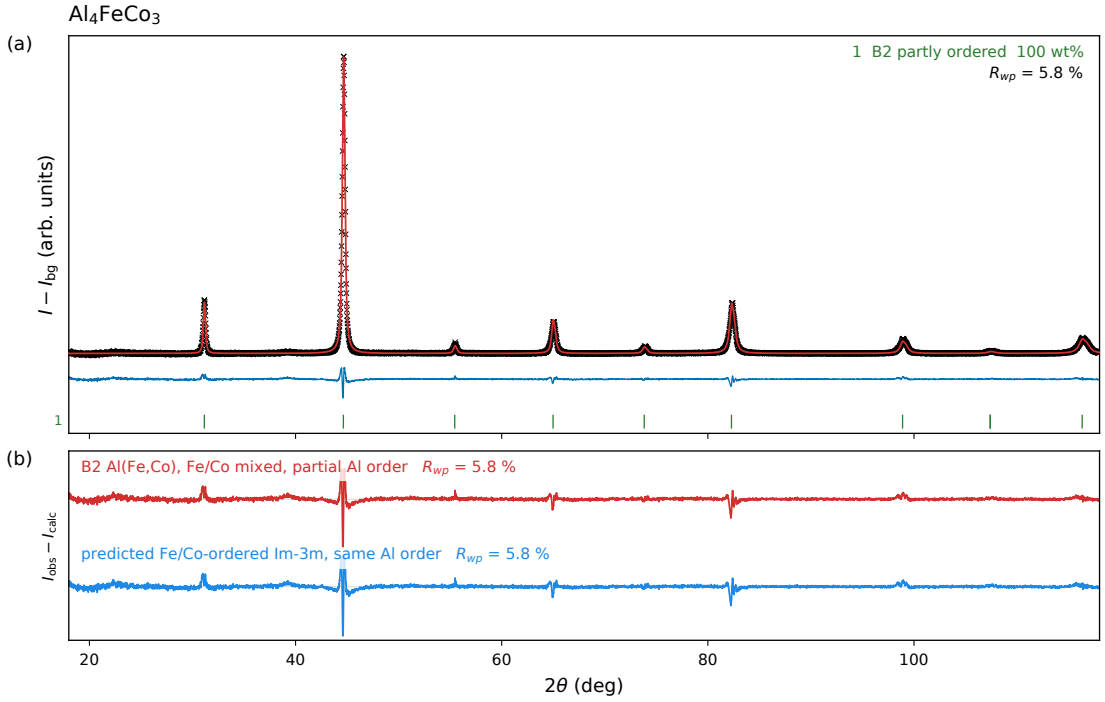

ScCrC2
(a)
1 ScCrC2 95 wt%
2 C 5 wt%
Rwp = 6.2 %
I − Ibg (arb. units)
1
2
(b)
P63/mmc ScCrC2 (Poettgen 1995) + graphite Rwp = 6.2 %
Sc/Cr-mixed cell (same cell) Rwp = 7.5 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
19.00
19.25
19.50
26.25
26.50
26.75
35.75
36.00
2θ (deg)

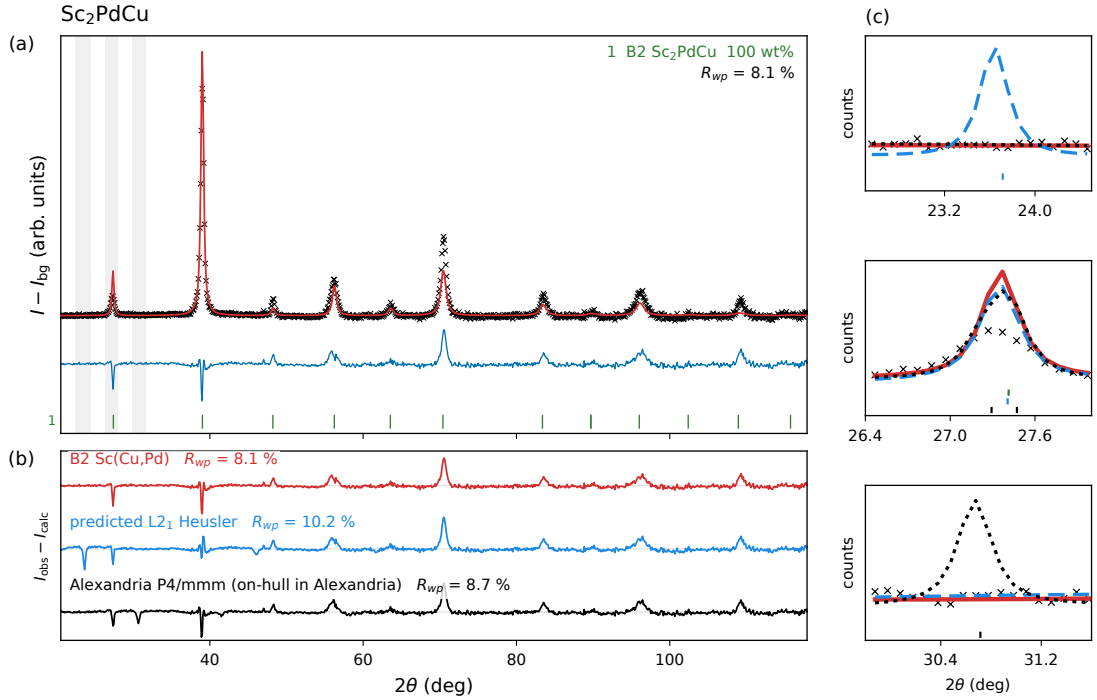

YCoSi
(a)
1 YCoSi 93 wt%
2 YCo2Si2 7 wt%
Rwp = 10.3 %
I − Ibg (arb. units)
1
2
(b)
Pnma YCoSi (known) + YCo2Si2 Rwp = 10.3 %
Iobs − Icalc
predicted P4/nmm alone Rwp = 54.8 %
20
40
60
80
100
120
140
2θ (deg)

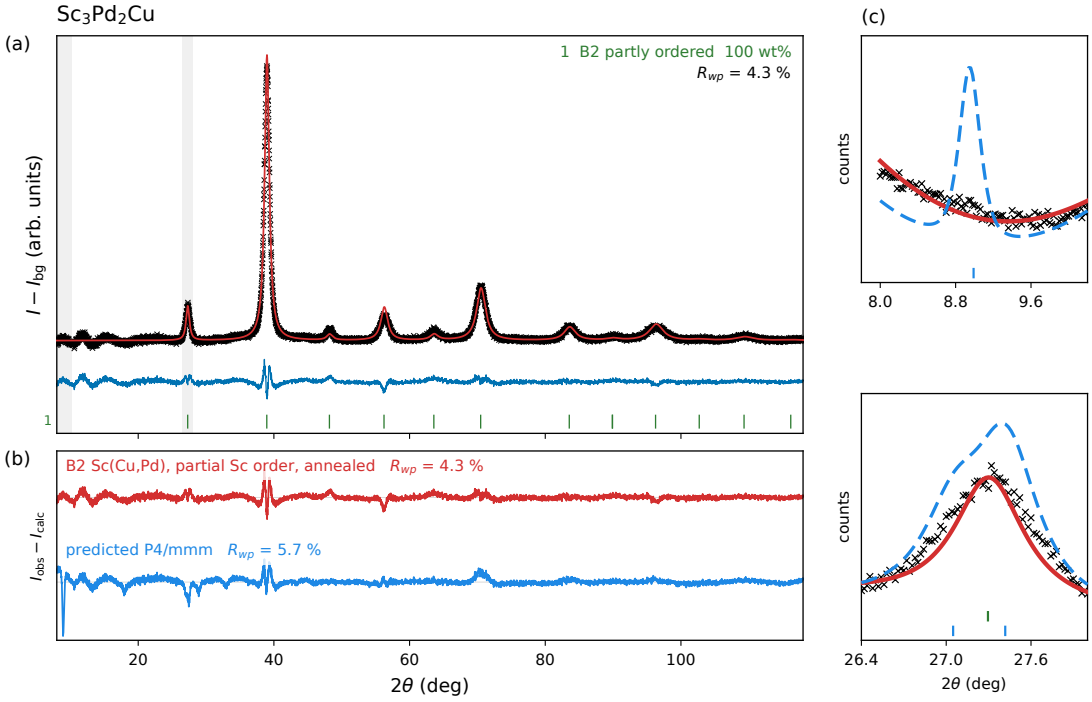

YCoSi3
(a)
1 YCoSi3 Cmmm (SmNiGe3-type) 100 wt%
Rwp = 12.9 %
I − Ibg (arb. units)
1
(b)
Cmmm YCoSi3 (SmNiGe3 type), single phase Rwp = 12.9 %
Iobs − Icalc
predicted I4mm alone Rwp = 53.6 %
20
40
60
80
100
120
140
2θ (deg)

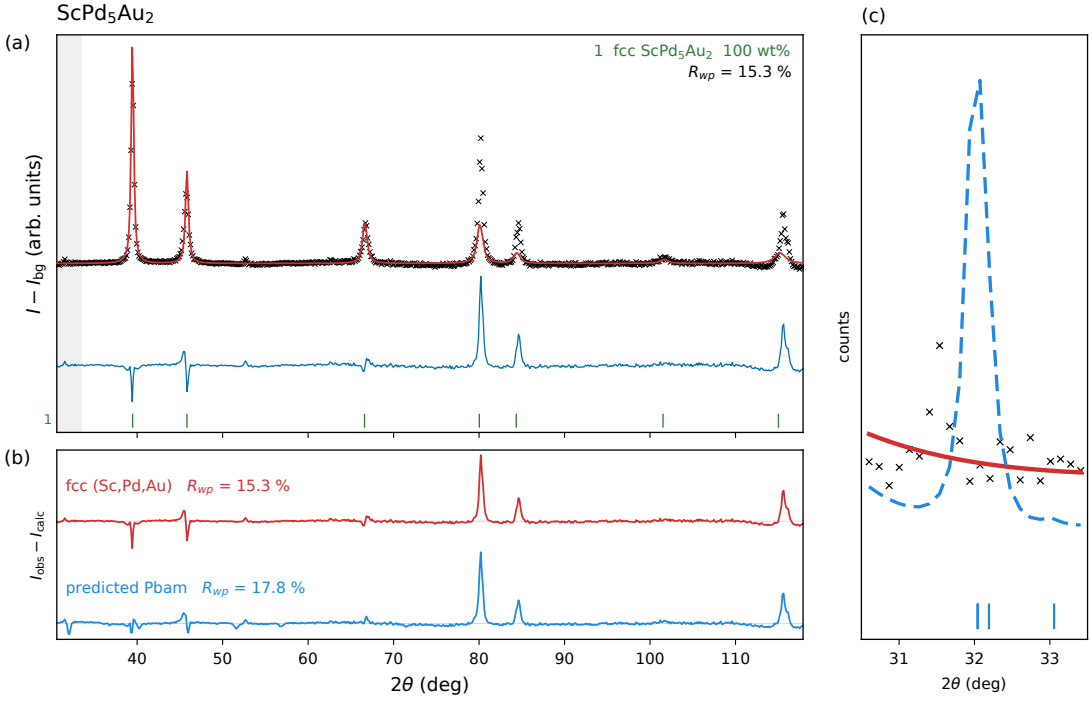

Al8FeCo7
(a)
1 B2 partly ordered 100 wt%
Rwp = 5.9 %
I − Ibg (arb. units)
1
(b)
B2 Al(Fe,Co), Fe/Co mixed, partial Al order Rwp = 5.9 %
predicted Fe/Co-ordered Pm-3m, same Al order Rwp = 5.9 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)

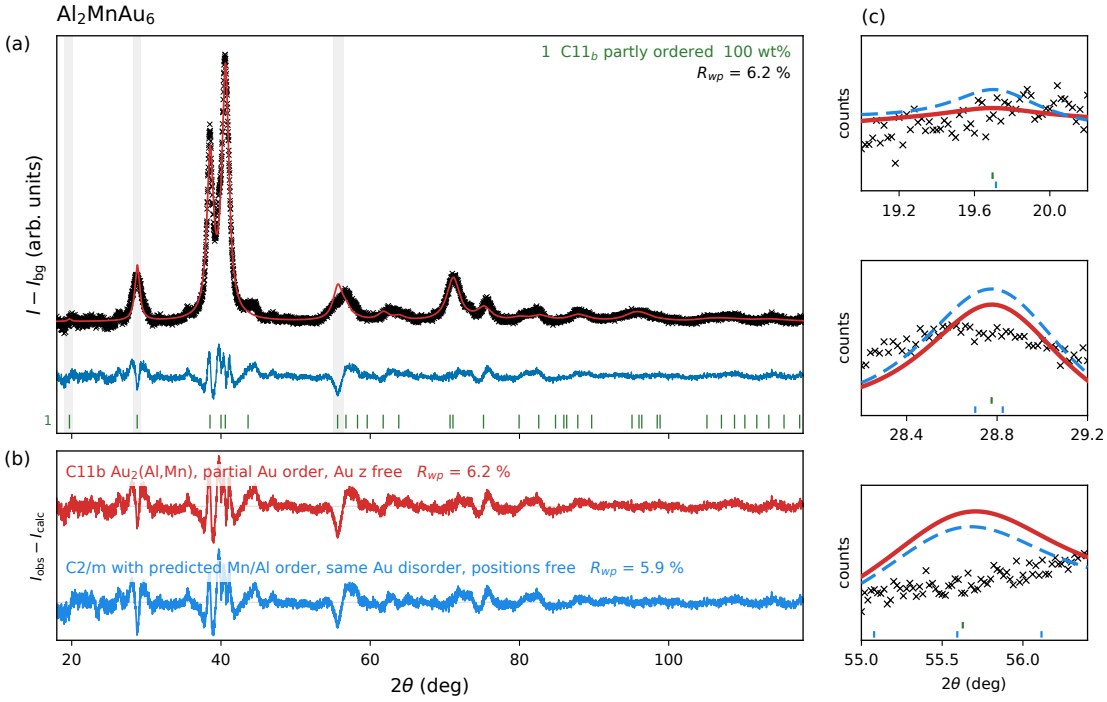

Hf2MoIr
(a)
1 B2 partly ordered 88 wt%
2 Hf2Ir E93 12 wt%
Rwp = 7.3 %
I − Ibg (arb. units)
1
2
(b)
bcc, partial Hf order + Hf2Ir Rwp = 7.3 %
predicted L21 Hf2MoIr + Hf2Ir Rwp = 16.9 %
Iobs − Icalc
20
40
60
80
100
120
140
2θ (deg)
(c)
counts
23.2
24.0
counts
27.0
27.5
28.0
2θ (deg)

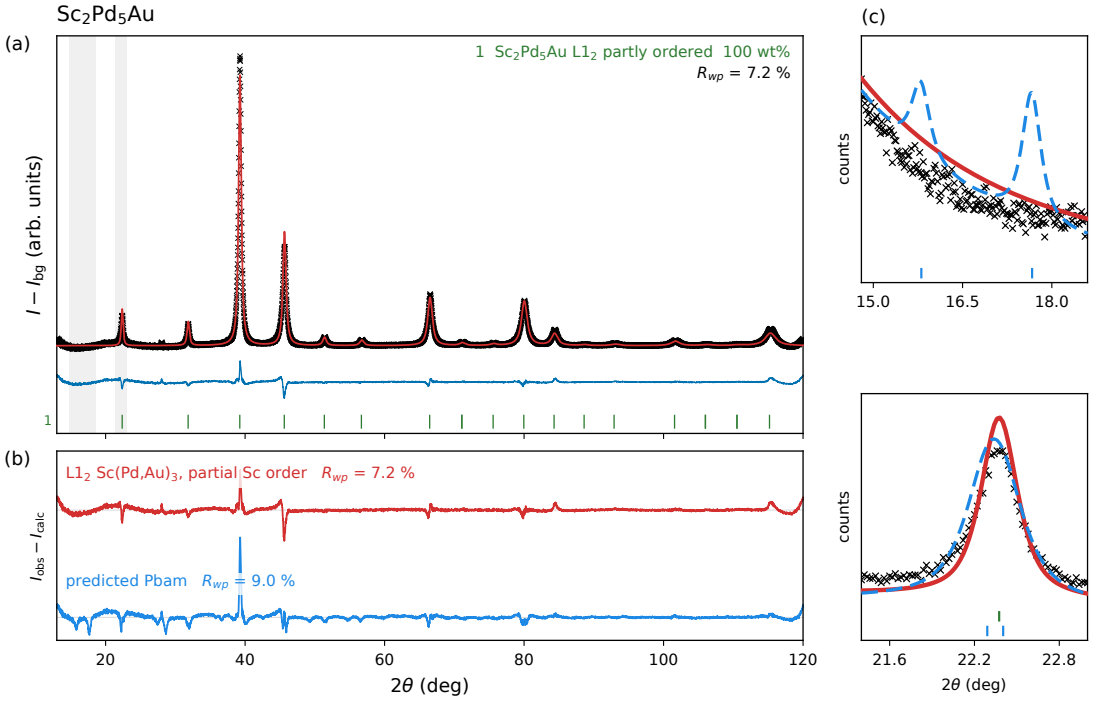

Al4FeCo3
(a)
1 B2 partly ordered 100 wt%
Rwp = 5.8 %
I − Ibg (arb. units)
1
(b)
B2 Al(Fe,Co), Fe/Co mixed, partial Al order Rwp = 5.8 %
predicted Fe/Co-ordered Im-3m, same Al order Rwp = 5.8 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)

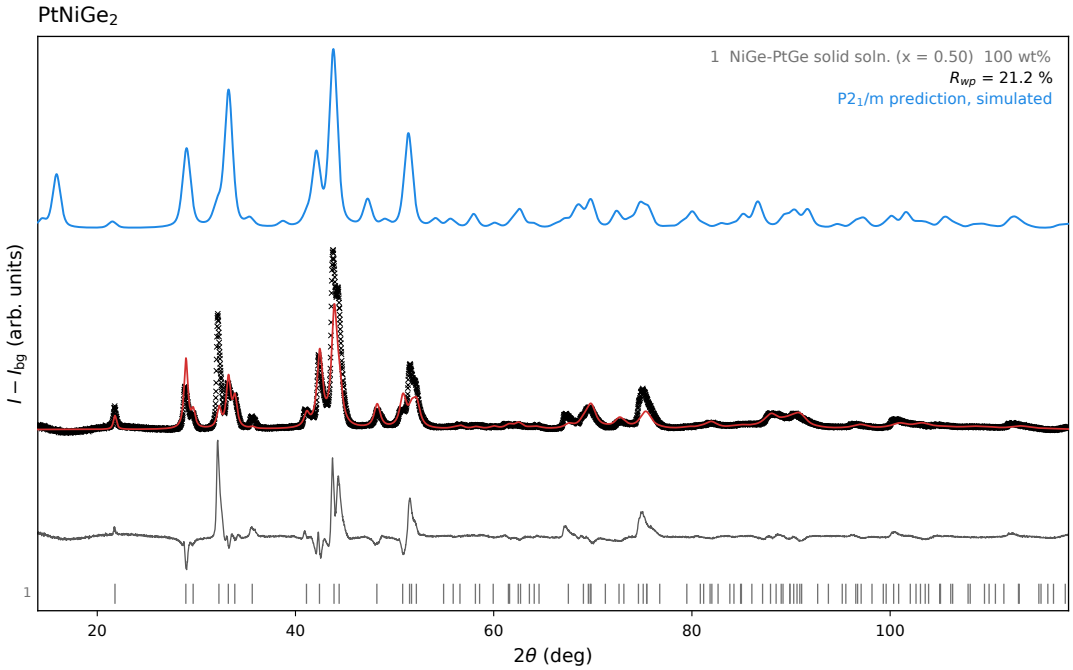

Sc2PdCu
(a)
1 B2 Sc2PdCu 100 wt%
Rwp = 8.1 %
I − Ibg (arb. units)
1
(b)
B2 Sc(Cu,Pd) Rwp = 8.1 %
predicted L21 Heusler Rwp = 10.2 %
Alexandria P4/mmm (on-hull in Alexandria) Rwp = 8.7 %
Iobs − Icalc
40
60
80
100
2θ (deg)
(c)
counts
23.2
24.0
26.4
27.0
27.6
30.4
31.2
2θ (deg)

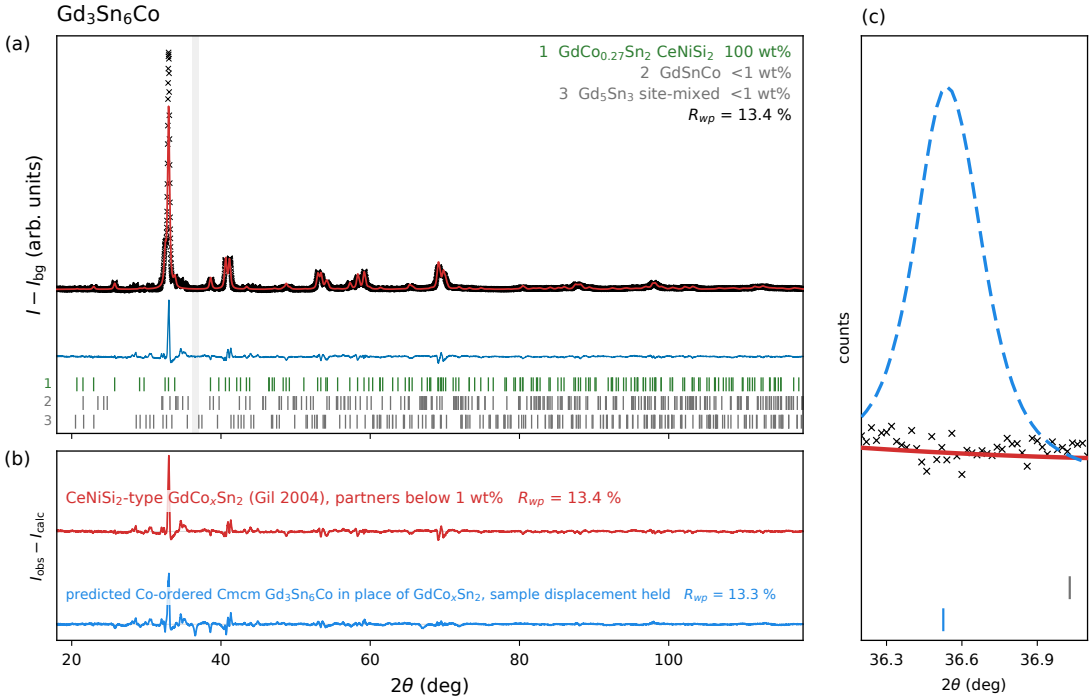

Sc3Pd2Cu
(a)
1 B2 partly ordered 100 wt%
Rwp = 4.3 %
I − Ibg (arb. units)
(b)
B2 Sc(Cu,Pd), partial Sc order, annealed Rwp = 4.3 %
predicted P4/mmm Rwp = 5.7 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)

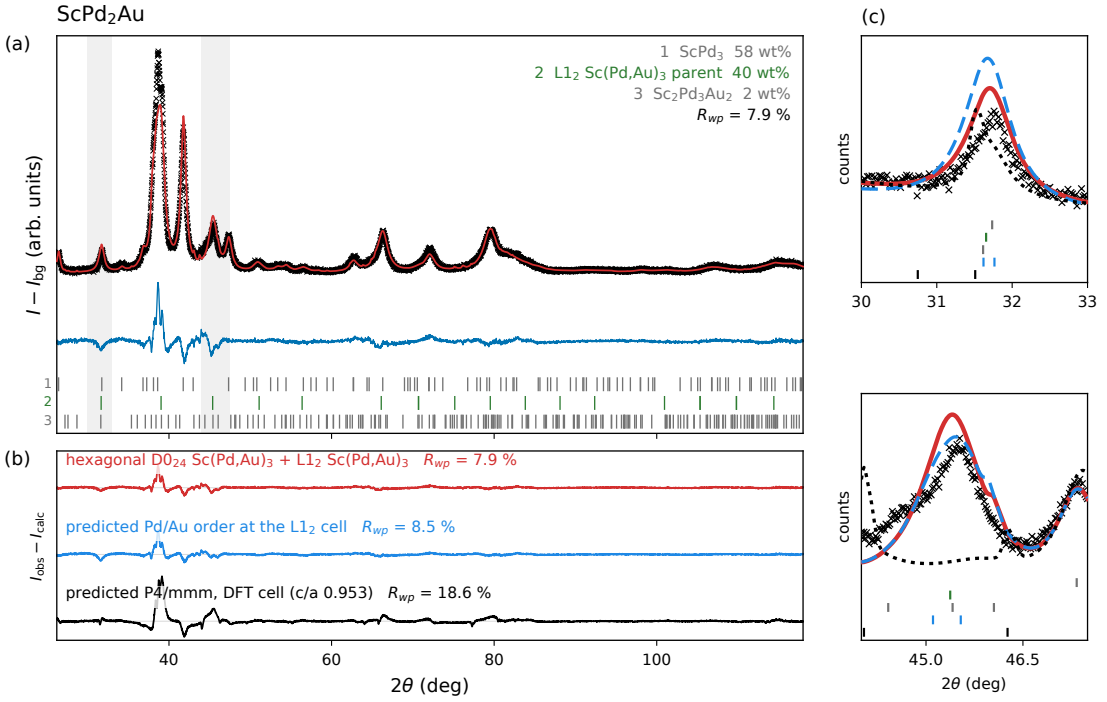

ScPd5Au2
(a)
1 fcc ScPd5Au2 100 wt%
Rwp = 15.3 %
I − Ibg (arb. units)
(b)
fcc (Sc,Pd,Au) Rwp = 15.3 %
predicted Pbam Rwp = 17.8 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)

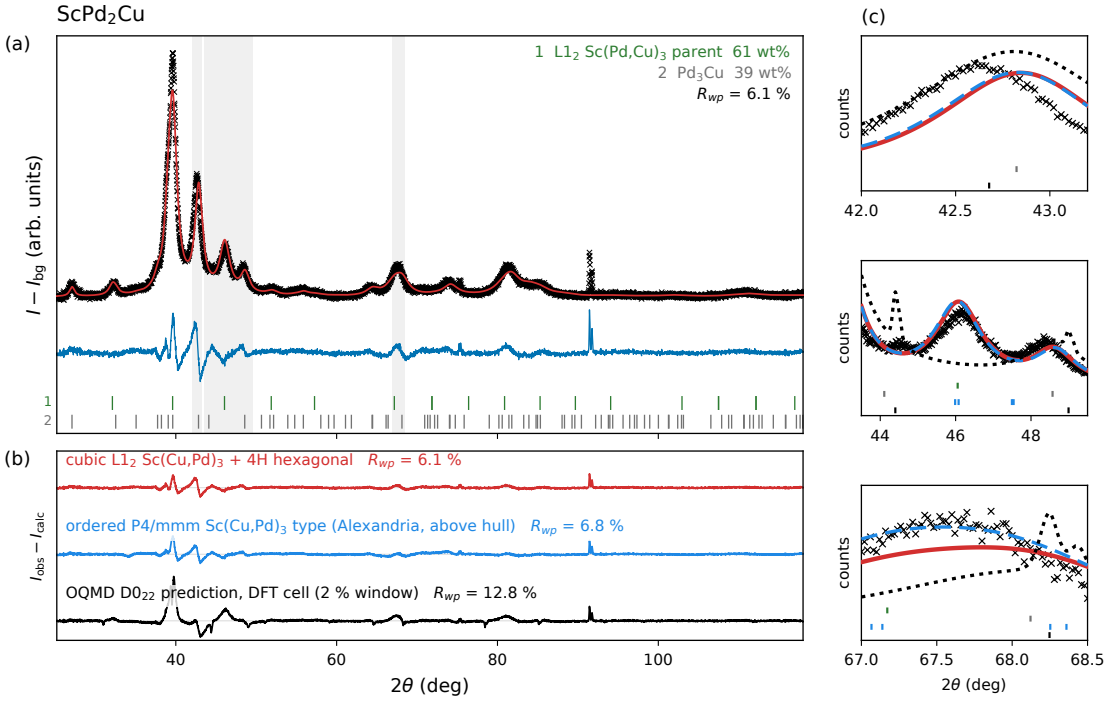

Al2MnAu6
(a)
1 C11b partly ordered 100 wt%
Rwp = 6.2 %
I − Ibg (arb. units)
(b)
C11b Au2(Al,Mn), partial Au order, Au z free Rwp = 6.2 %
C2/m with predicted Mn/Al order, same Au disorder, positions free Rwp = 5.9 %
Iobs − Icalc
2θ (deg)
(c)
counts


Sc2Pd5Au
(a)
1 Sc2Pd5Au L12 partly ordered 100 wt%
Rwp = 7.2 %
I − Ibg (arb. units)
(b)
L12 Sc(Pd,Au)3, partial Sc order Rwp = 7.2 %
predicted Pbam Rwp = 9.0 %
Iobs − Icalc
2θ (deg)
(c)
counts

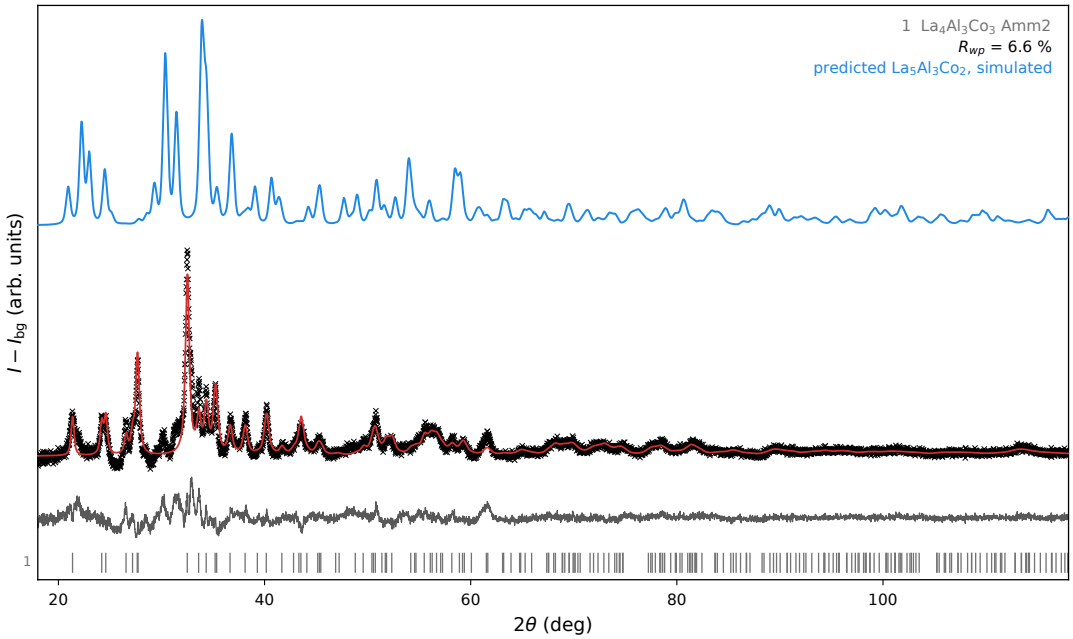

PtNiGe2
1 NiGe-PtGe solid soln. (x = 0.50) 100 wt%
Rwp = 21.2 %
P21/m prediction, simulated
I − Ibg (arb. units)
1
20
40
60
80
100
2θ (deg)


Gd3Sn6Co
(a)
1 GdCo0.27Sn2 CeNiSi2 100 wt%
2 GdSnCo <1 wt%
3 Gd5Sn3 site-mixed <1 wt%
Rwp = 13.4 %
I − Ibg (arb. units)
1
2
3
(b)
CeNiSi2-type GdCoxSn2 (Gil 2004), partners below 1 wt% Rwp = 13.4 %
predicted Co-ordered Cmcm Gd3Sn6Co in place of GdCoxSn2, sample displacement held Rwp = 13.3 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
36.3
36.6
36.9
2θ (deg)

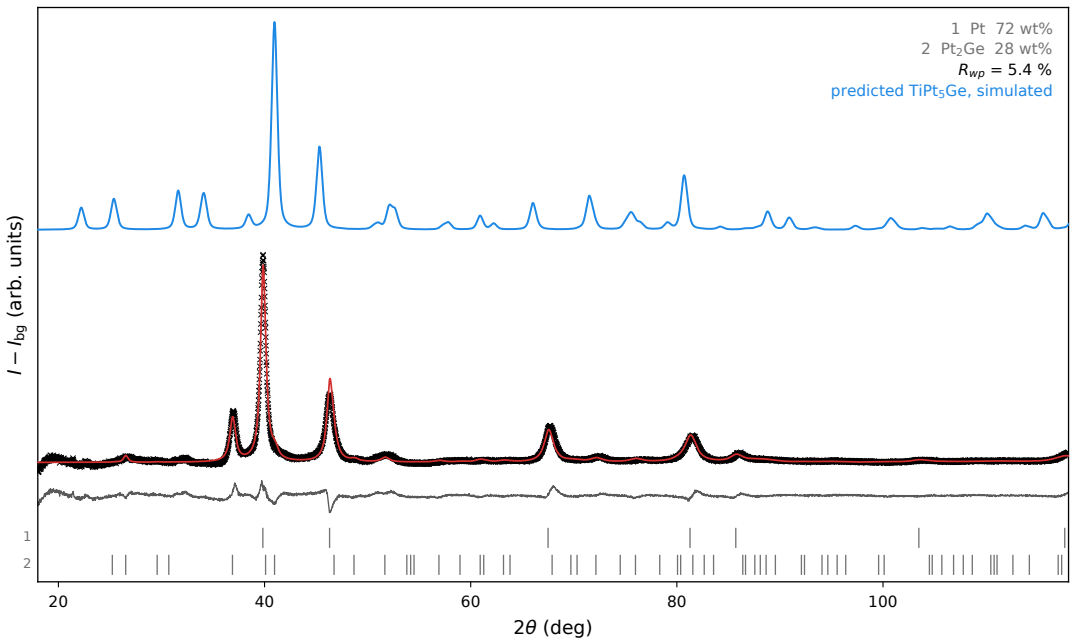

ScPd2Au
(a)
1 ScPd3 58 wt%
2 L12 Sc(Pd,Au)3 parent 40 wt%
3 Sc2Pd3Au2 2 wt%
Rwp = 7.9 %
I − Ibg (arb. units)
(b)
hexagonal D024 Sc(Pd,Au)3 + L12 Sc(Pd,Au)3 Rwp = 7.9 %
predicted Pd/Au order at the L12 cell Rwp = 8.5 %
predicted P4/mmm, DFT cell (c/a 0.953) Rwp = 18.6 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)

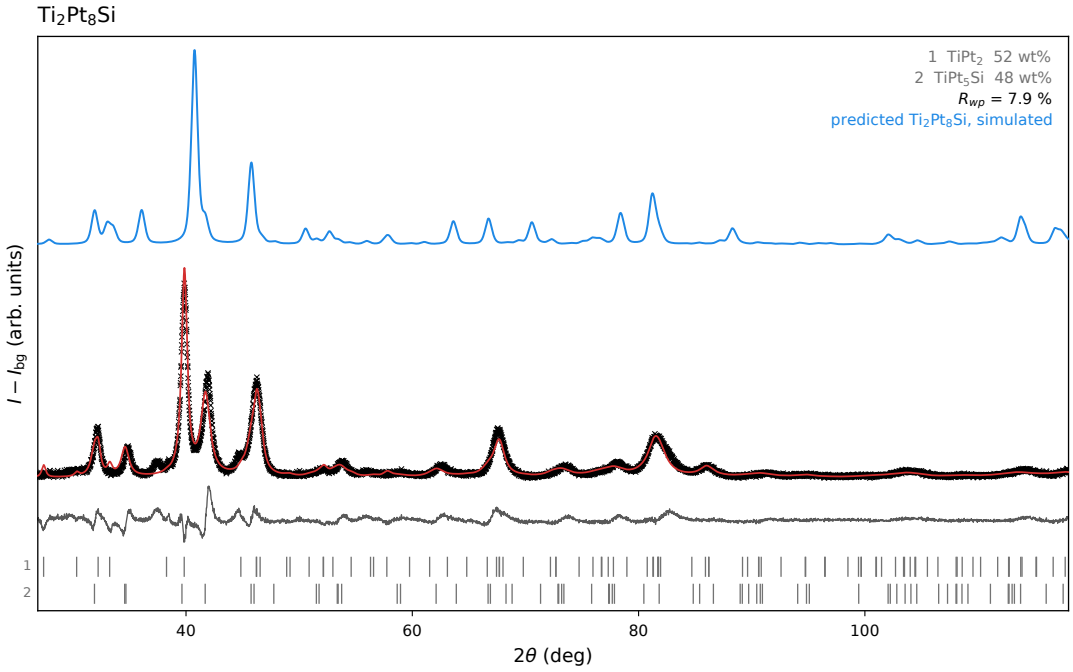

ScPd2Cu
(a)
1 L12 Sc(Pd,Cu)3 parent 61 wt%
2 Pd3Cu 39 wt%
Rwp = 6.1 %
I − Ibg (arb. units)
(b)
cubic L12 Sc(Cu,Pd)3 + 4H hexagonal Rwp = 6.1 %
ordered P4/mmm Sc(Cu,Pd)3 type (Alexandria, above hull) Rwp = 6.8 %
OQMD D022 prediction, DFT cell (2 % window) Rwp = 12.8 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)

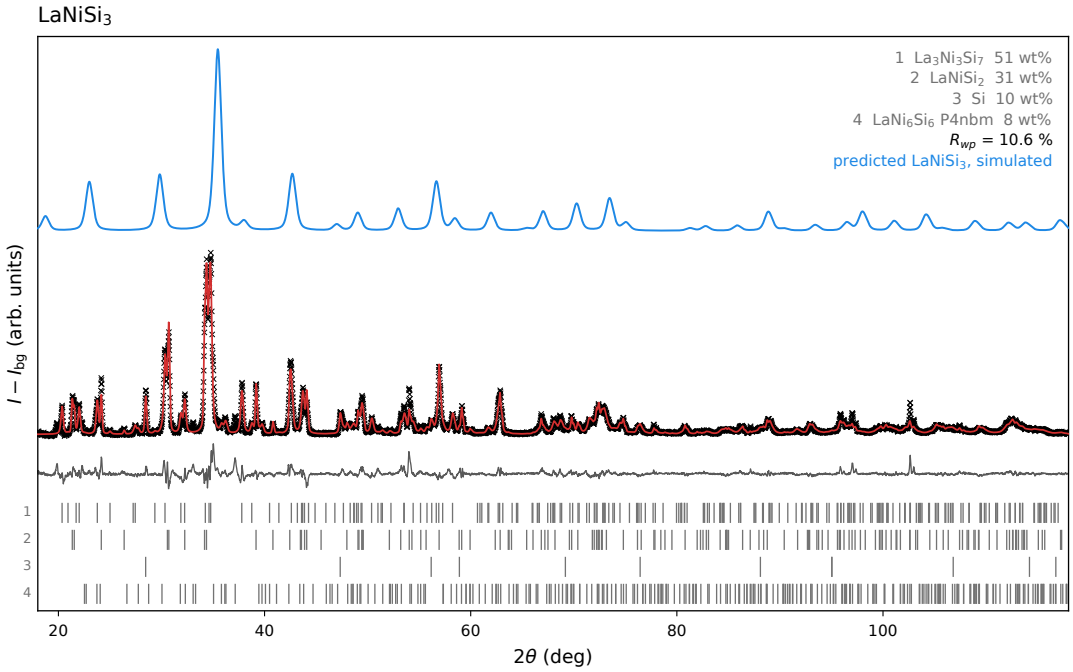
$La_5Al_3Co_2$

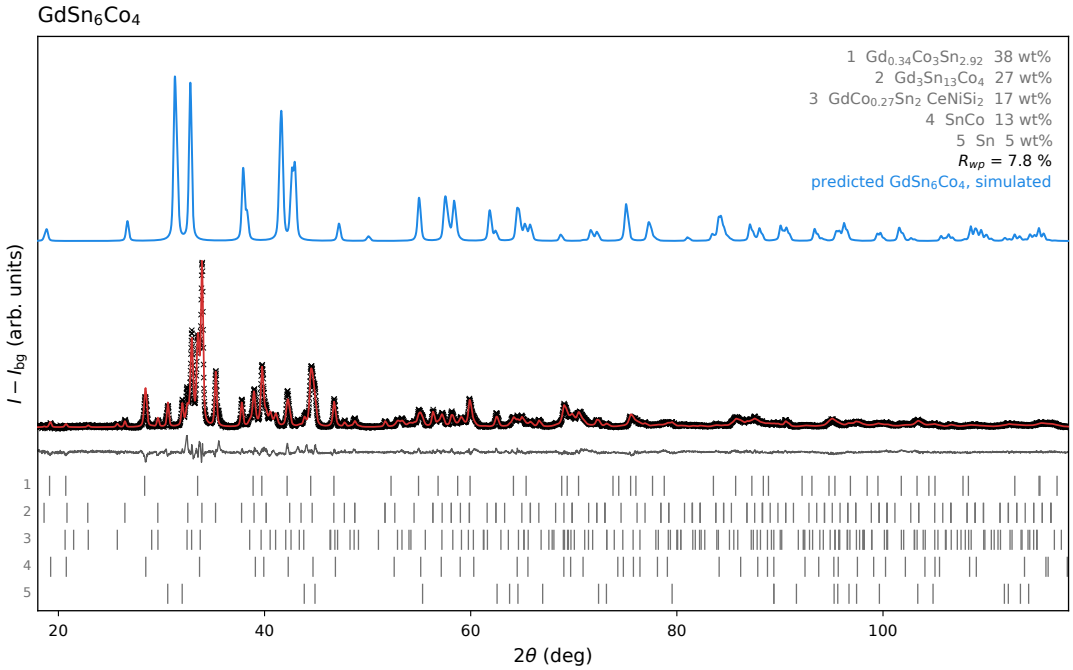


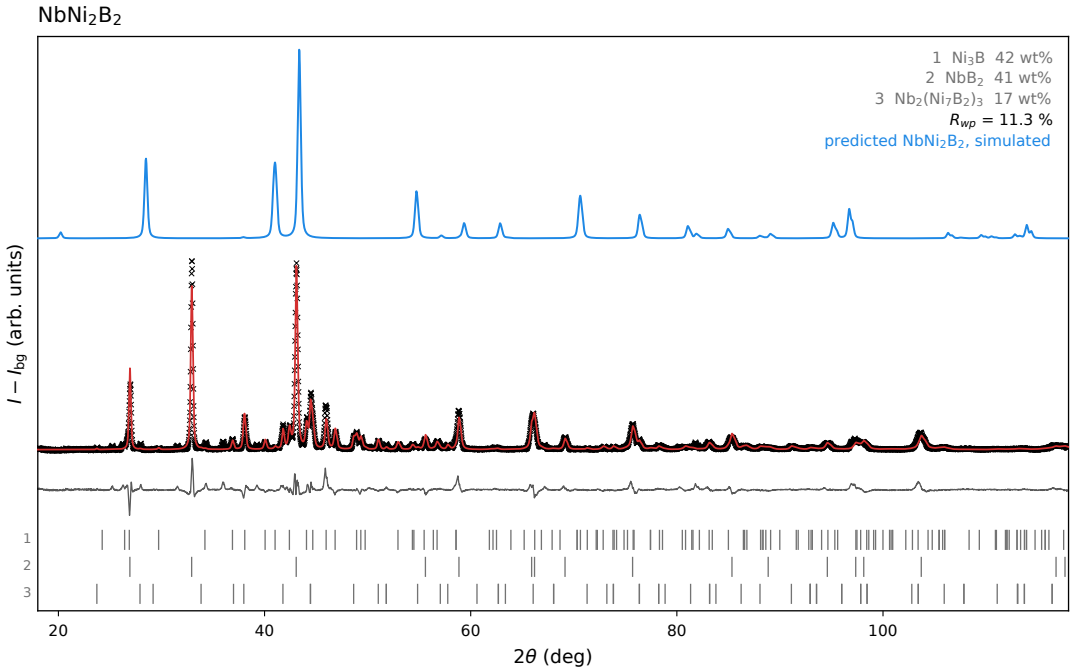
$TiPt_5Ge$



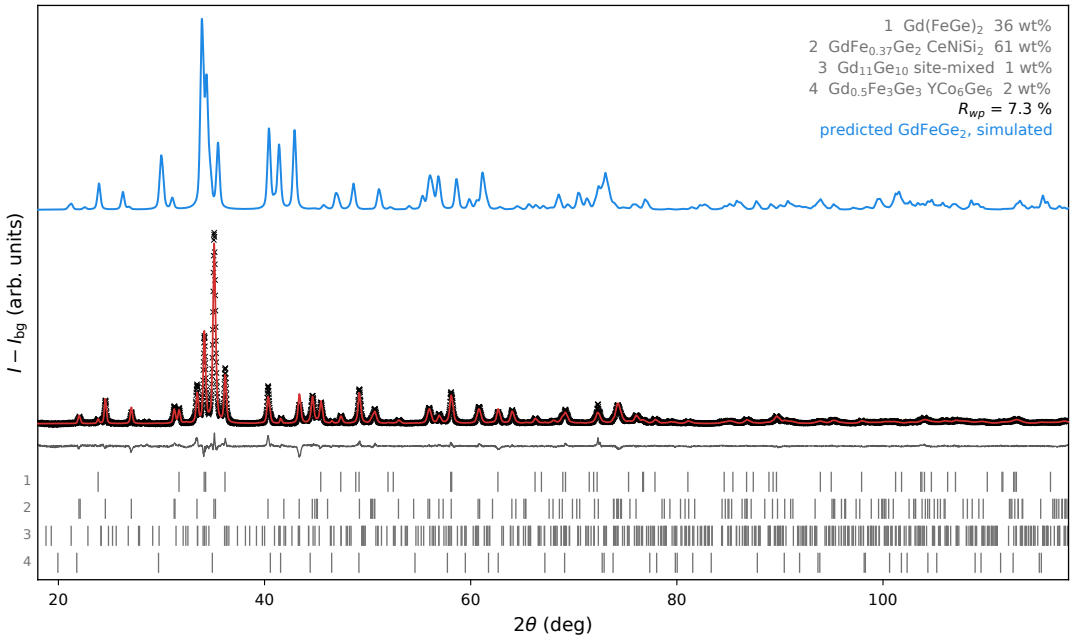

Ti2Pt8Si
1 TiPt2 52 wt%
2 TiPt5Si 48 wt%
Rwp = 7.9 %
predicted Ti2Pt8Si, simulated
I − Ibg (arb. units)
1
2
40
60
80
100
2θ (deg)


LaNiSi3
1 La3Ni3Si7 51 wt%
2 LaNiSi2 31 wt%
3 Si 10 wt%
4 LaNi6Si6 P4nbm 8 wt%
Rwp = 10.6 %
predicted LaNiSi3, simulated
I − Ibg (arb. units)
1
2
3
4
20
40
60
80
100
2θ (deg)

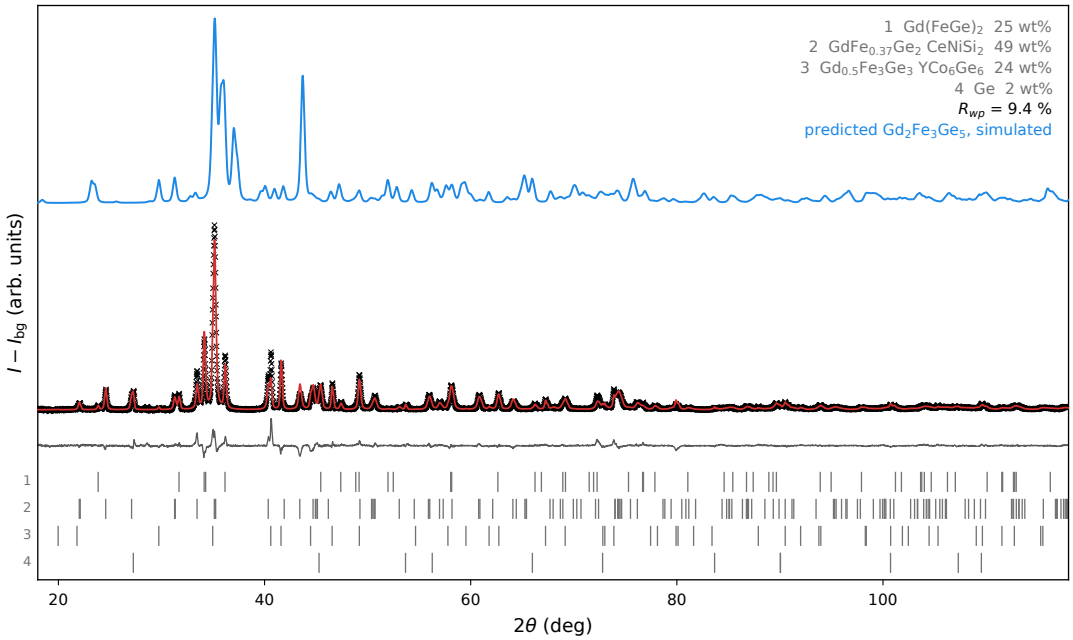

GdSn6Co4
1 Gd0.34Co3Sn2.92 38 wt%
2 Gd3Sn13Co4 27 wt%
3 GdCo0.27Sn2 CeNiSi2 17 wt%
4 SnCo 13 wt%
5 Sn 5 wt%
Rwp = 7.8 %
predicted GdSn6Co4, simulated
I − Ibg (arb. units)
1
2
3
4
5
20
40
60
80
100
2θ (deg)


NbNi2B2
1 Ni3B 42 wt%
2 NbB2 41 wt%
3 Nb2(Ni7B2)3 17 wt%
Rwp = 11.3 %
predicted NbNi2B2, simulated
I − Ibg (arb. units)
1
2
3
20
40
60
80
100
2θ (deg)

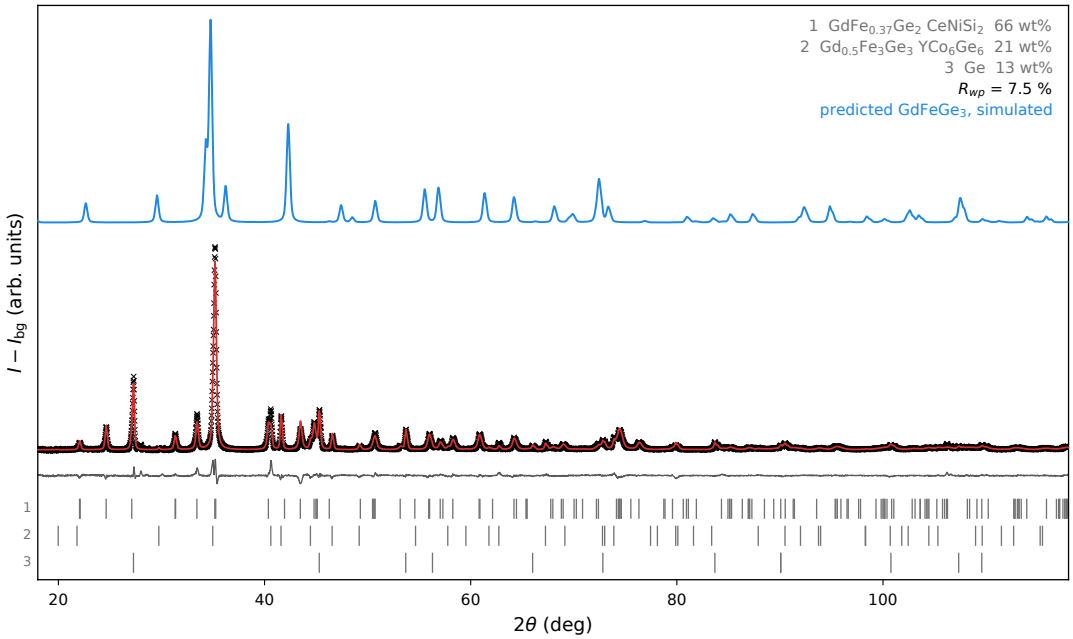
GdFeGe$_2$



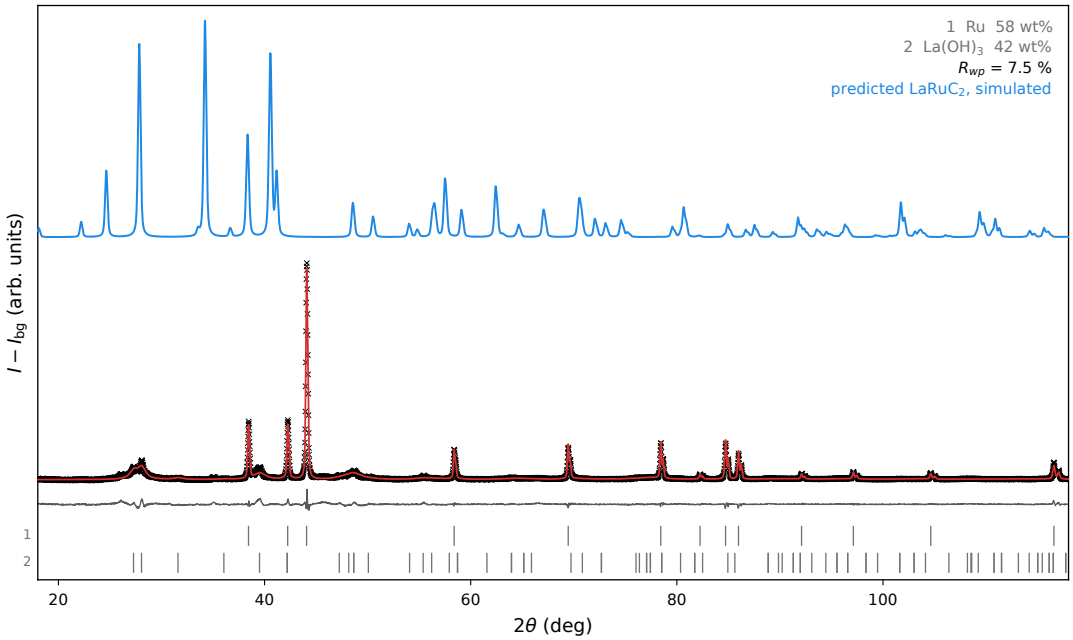
Gd$_2$Fe$_3$Ge$_5$

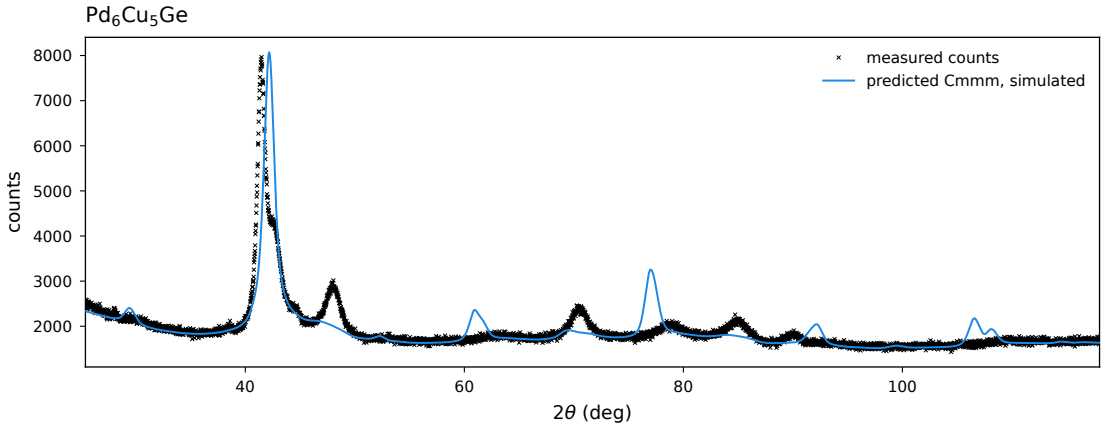


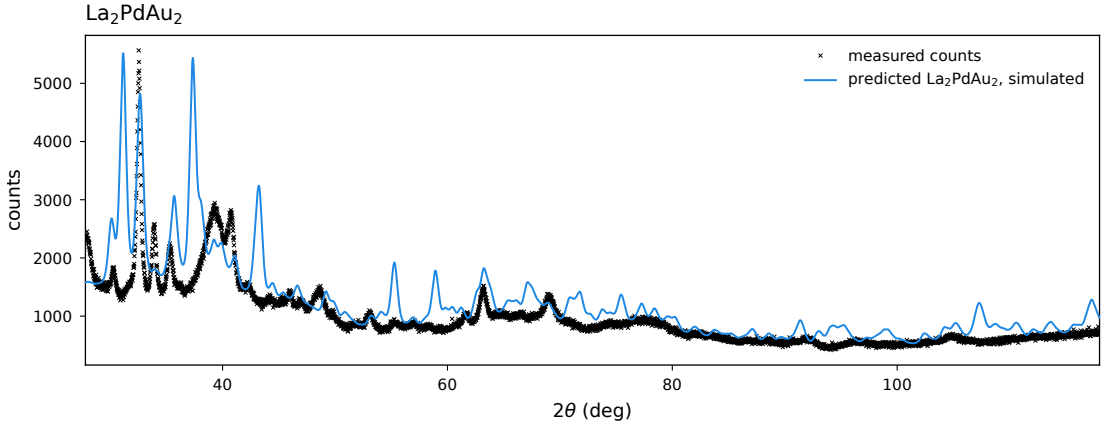
GdFeGe$_3$

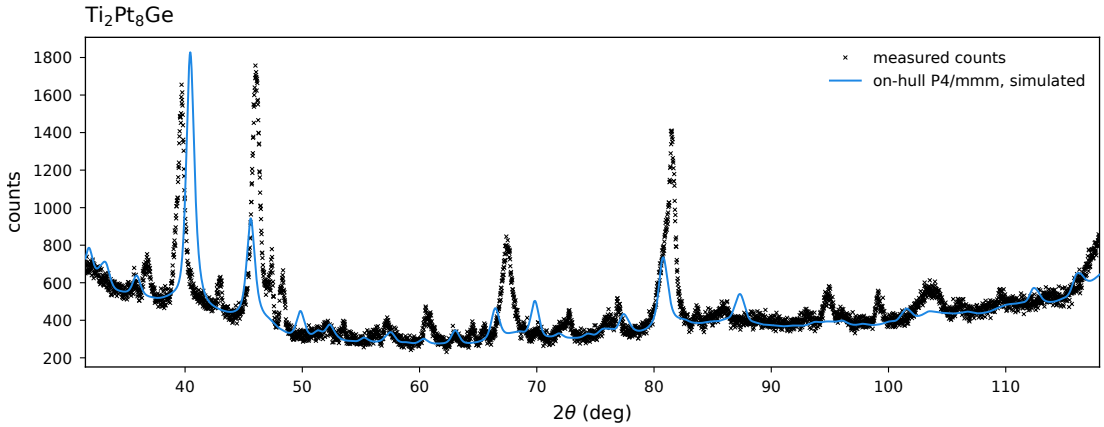


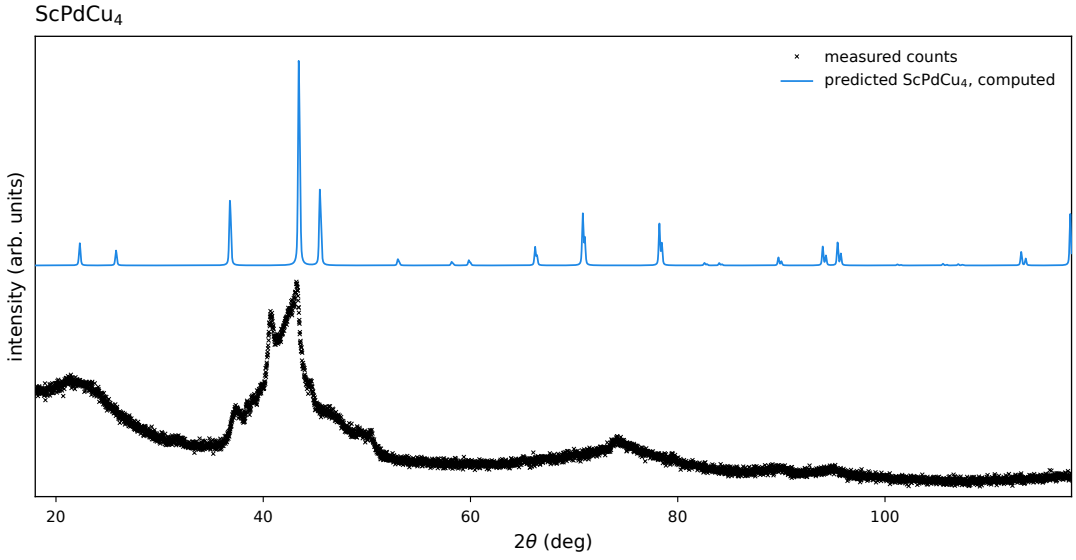
LaRuC$_2$

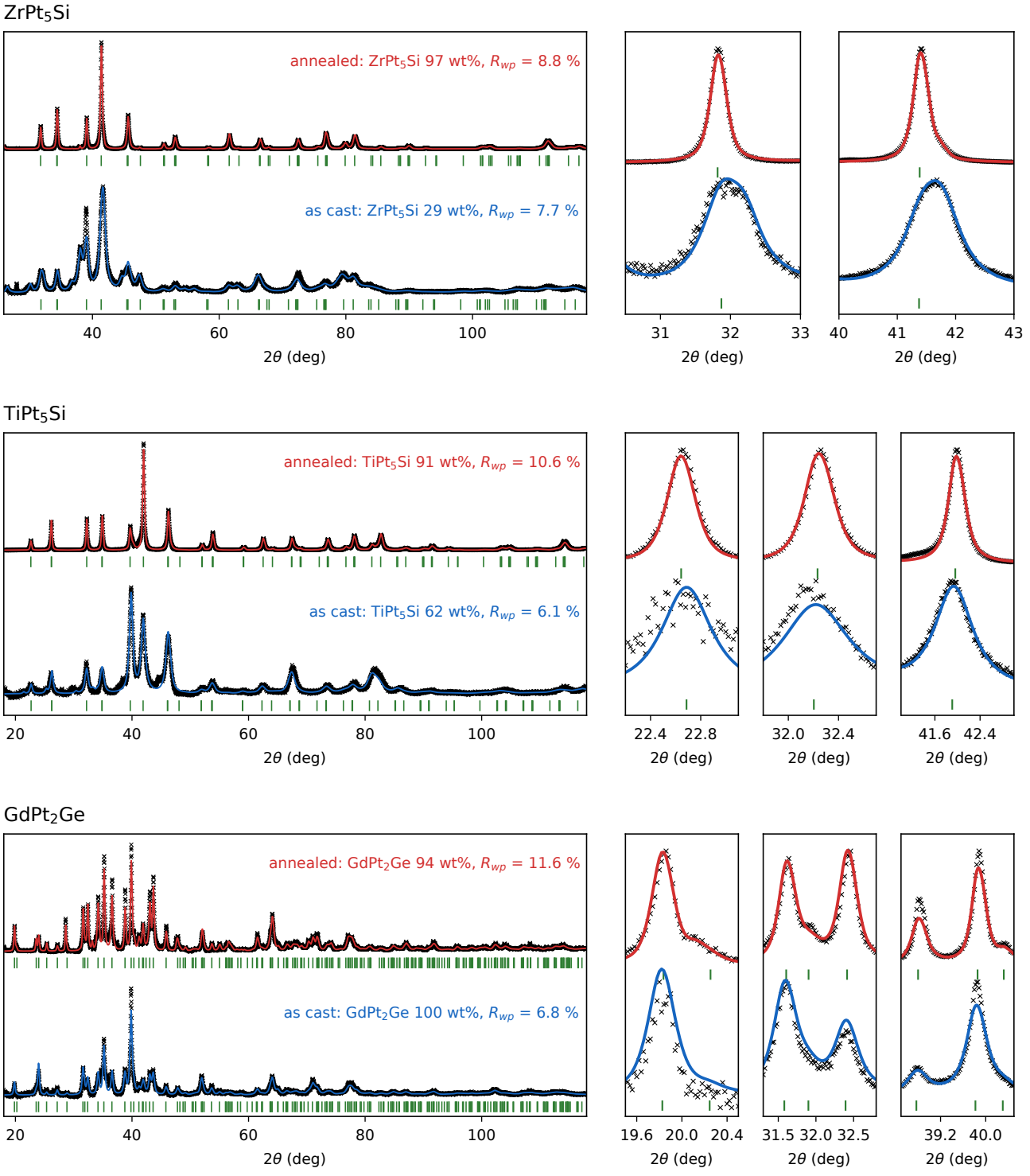



Pd6Cu5Ge
measured counts
predicted Cmmm, simulated
counts
2θ (deg)


La2PdAu2
measured counts
predicted La2PdAu2, simulated
counts
2θ (deg)


Ti2Pt8Ge
measured counts
on-hull P4/mmm, simulated
counts
2θ (deg)


ScPdCu4
measured counts
predicted ScPdCu4, computed
intensity (arb. units)
2θ (deg)


ZrPt5Si
annealed: ZrPt5Si 97 wt%, Rwp = 8.8 %
as cast: ZrPt5Si 29 wt%, Rwp = 7.7 %
40
60
80
100
2θ (deg)
31
32
33
2θ (deg)
40
41
42
43
2θ (deg)
TiPt5Si
annealed: TiPt5Si 91 wt%, Rwp = 10.6 %
as cast: TiPt5Si 62 wt%, Rwp = 6.1 %
20
40
60
80
100
2θ (deg)
22.4
22.8
2θ (deg)
32.0
32.4
2θ (deg)
41.6
42.4
2θ (deg)
GdPt2Ge
annealed: GdPt2Ge 94 wt%, Rwp = 11.6 %
as cast: GdPt2Ge 100 wt%, Rwp = 6.8 %
20
40
60
80
100
2θ (deg)
19.6
20.0
20.4
2θ (deg)
31.5
32.0
32.5
2θ (deg)
39.2
40.0
2θ (deg)